Question 3.2 – Bernoulli Estimate at Junction

Applying Bernoulli's equation along a streamline from the inlet face to the junction stagnation point:

$$P_1 + \frac{1}{2}\rho V_1^2 + \rho g z_1 = P_2 + \frac{1}{2}\rho V_2^2 + \rho g z_2$$

Assumptions:

- Fluid is water: $\rho = 1000$ kg/m^3
- Inlet velocity: $V_1 = 1.353$ m/s
- Inlet pressure (gauge): $P_1 = 0$ Pa
- Stagnation point: $V_2 = 0$ m/s
- Neglect elevation change: $z_1 \approx z_2$

Bernoulli's equation simplifies to:

$$P_2 = P_1 + \frac{1}{2}\rho V_1^2$$

$$P_2 = 0 + \frac{1}{2}(1000)(1.353)^2$$

$$\boxed{P_2 \approx 915 \text{ Pa}}$$